Year 7

REVISION PACK

Unit 3

Name: _____

Class: _____

Unit 3 · Collecting like terms

Worked example: simplifying expressions

Simplify, by collecting like terms.

(a) $6x + 2x - x$

(b) $-3p + 7p - 2p$

(c) $5x + 4y - 2x - y$

(d) $a^2 + 4a + 2a^2 - a$

(a)	**Step 1**	Add and subtract the numbers	$6 + 2 - 1 = 7$
	Step 2	Answer	$7x$
(b)	**Step 1**	Add and subtract the numbers	$-3 + 7 - 2 = 2$
	Step 2	Answer	$2p$
(c)	**Step 1**	Collect the x terms	$5x - 2x = 3x$
	Step 2	Collect the y terms	$4y - y = 3y$
	Step 3	Answer	$3x + 3y$
(d)	**Step 1**	Collect the a^2 terms	$a^2 + 2a^2 = 3a^2$
	Step 2	Collect the a terms	$4a - a = 3a$
	Step 3	Answer	$3a^2 + 3a$

Practice: simplifying expressions

1.1 Simplify, by collecting like terms.

(a) $3x + 7x - x$

(b) $-4p + 9p - 2p$

(c) $6x + 2y - x - 5y$

(d) $r^2 + 6r + 4r^2 - 2r$

1.2 Simplify, by collecting like terms.

(a) $5y + 2y - 3y$

(b) $-2q + 8q - 3q$

(c) $7x + 5y - 3x - 2y$

(d) $m^2 + 3m + m^2 - m$

1.3 Simplify, by collecting like terms.

(a) $8a + 4a - 5a$

(b) $-5b + 11b - 4b$

(c) $9x + 3y - 4x - y$

(d) $n^2 + 7n + 5n^2 - 4n$

1.4 Simplify, by collecting like terms.

(a) $2m + 9m - 6m$

(b) $-3k + 10k - 6k$

(c) $5x + 6y - 2x - 4y$

(d) $t^2 + 2t + 2t^2 - t$

1.5 Simplify, by collecting like terms.

(a) $10c + 3c - 7c$

(b) $-6d + 12d - d$

(c) $8x + 9y - 6x - 3y$

(d) $x^2 + 8x + 6x^2 - 3x$

1.6 Simplify, by collecting like terms.

(a) $6z + 6z - 4z$

(b) $-w + 7w - 2w$

(c) $4x + 7y - 3x - 6y$

(d) $y^2 + 5y + y^2 - 2y$

Unit 3 · Writing expressions

Worked example: writing expressions from words

(a) Sara is h cm tall. Omar is 6 cm shorter than Sara. Write an expression for Omar's height.

(b) A bag of rice has a mass of 5 kg. Write an expression for the mass of n bags in kg and in grams.

(c) Books cost x dollars each. Pens cost y dollars each. Write an expression for T, the cost of 4 books and 3 pens.

(d) AB = BC = x cm. CD is 2 times as long as AB. DE is 5 cm. Write an expression for the length AE.

(a)	Step 1	Shorter: subtract	$h - 6$
	Step 2	Answer	$h - 6$ cm
(b)	Step 1	n bags: times by n	$5n$ kg
	Step 2	1 kg = 1000 g	$5n \times 1000 = 5000n$ g
(c)	Step 1	4 books, 3 pens	$4x + 3y$
	Step 2	Answer	$T = 4x + 3y$
(d)	Step 1	Add all the parts	$x + x + 2x + 5$
	Step 2	Answer	$4x + 5$ cm

Practice: writing expressions from words

2.1

(a) Bahir is h cm tall. Sienna is 4 cm taller than Bahir. Write an expression for Sienna's height.

(b) A bag of sugar has a mass of 2 kg. Write an expression for the mass of n bags in kg and in grams.

(c) Pens cost x pence each. Rulers cost y pence each. Write an expression for T, the cost of 3 pens and 2 rulers.

(d) AB = BC = x cm. CD is 2 times as long as AB. DE is 3 cm. Write an expression for the length AE.

2.2

(a) Ali is h cm tall. Mona is 9 cm shorter than Ali. Write an expression for Mona's height.

(b) A bag of flour has a mass of 3 kg. Write an expression for the mass of n bags in kg and in grams.

(c) Apples cost x pence each. Oranges cost y pence each. Write an expression for T, the cost of 5 apples and 4 oranges.

(d) AB = BC = x cm. CD is 3 times as long as AB. DE is 7 cm. Write an expression for the length AE.

2.3

(a) Rana is h cm tall. Yusuf is 12 cm taller than Rana. Write an expression for Yusuf's height.

(b) A bag of cement has a mass of 4 kg. Write an expression for the mass of n bags in kg and in grams.

(c) Pencils cost x pence each. Rubbers cost y pence each. Write an expression for T, the cost of 2 pencils and 6 rubbers.

(d) AB = BC = x cm. CD is 4 times as long as AB. DE is 2 cm. Write an expression for the length AE.

Practice: writing expressions from words (continued)

2.4

(a) Hadi is h cm tall. Lina is 7 cm shorter than Hadi. Write an expression for Lina's height.

(b) A bag of potatoes has a mass of 10 kg. Write an expression for the mass of n bags in kg and in grams.

(c) Cakes cost x pence each. Drinks cost y pence each. Write an expression for T, the cost of 6 cakes and 2 drinks.

(d) AB = BC = x cm. CD is 2 times as long as AB. DE is 10 cm. Write an expression for the length AE.

2.5

(a) Nour is h cm tall. Zaid is 15 cm taller than Nour. Write an expression for Zaid's height.

(b) A bag of sand has a mass of 6 kg. Write an expression for the mass of n bags in kg and in grams.

(c) Stamps cost x pence each. Cards cost y pence each. Write an expression for T, the cost of 4 stamps and 5 cards.

(d) AB = BC = x cm. CD is 5 times as long as AB. DE is 1 cm. Write an expression for the length AE.

2.6

(a) Amira is h cm tall. Tariq is 3 cm shorter than Amira. Write an expression for Tariq's height.

(b) A bag of rice has a mass of 25 kg. Write an expression for the mass of n bags in kg and in grams.

(c) Cups cost x pence each. Plates cost y pence each. Write an expression for T, the cost of 7 cups and 3 plates.

(d) AB = BC = x cm. CD is 3 times as long as AB. DE is 8 cm. Write an expression for the length AE.

Unit 3 · Substitution

Worked example: substituting into formulae

$y = mx + c$. Find y when

(a) $m = 5$, $x = 2$, $c = 3$

(b) $m = -3$, $x = -4$, $c = -2$

(c) $T = RH$. Find T when $R = 12$, $H = 6$

(d) $A = x^2 - 7$. Find A when $x = 5$

(a)	**Step 1**	Put the numbers in	$5 \times 2 + 3 = 10 + 3 = 13$
(b)	**Step 1**	Put the numbers in	$-3 \times (-4) + (-2) = 12 - 2 = 10$
(c)	**Step 1**	RH means $R \times H$	$12 \times 6 = 72$
(d)	**Step 1**	Square first	$5^2 - 7 = 25 - 7 = 18$

Practice: substituting into formulae

3.1 $y = mx + c$. Find y when

(a) $m = 8$, $x = 3$, $c = 4$

(b) $m = -2$, $x = -5$, $c = -5$

(c) $T = RH$. Find T when $R = 10$, $H = 40$

(d) $A = x^2 - 5$. Find A when $x = 4$

3.2 $y = mx + c$. Find y when

(a) $m = 6$, $x = 4$, $c = 7$

(b) $m = -4$, $x = -3$, $c = -6$

(c) $T = RH$. Find T when $R = 15$, $H = 8$

(d) $A = x^2 - 9$. Find A when $x = 6$

3.3 $y = mx + c$. Find y when

(a) $m = 3$, $x = 9$, $c = 2$

(b) $m = -5$, $x = -2$, $c = -3$

(c) $T = RH$. Find T when $R = 11$, $H = 7$

(d) $A = x^2 - 10$. Find A when $x = 7$

3.4 $y = mx + c$. Find y when

(a) $m = 7$, $x = 5$, $c = 6$

(b) $m = -3$, $x = -6$, $c = -8$

(c) $T = RH$. Find T when $R = 9$, $H = 30$

(d) $A = x^2 - 4$. Find A when $x = 3$

3.5 $y = mx + c$. Find y when

(a) $m = 4$, $x = 8$, $c = 9$

(b) $m = -6$, $x = -1$, $c = -4$

(c) $T = RH$. Find T when $R = 20$, $H = 12$

(d) $A = x^2 - 12$. Find A when $x = 8$

3.6 $y = mx + c$. Find y when

(a) $m = 9$, $x = 2$, $c = 5$

(b) $m = -2$, $x = -7$, $c = -9$

(c) $T = RH$. Find T when $R = 14$, $H = 5$

(d) $A = x^2 - 1$. Find A when $x = 10$

Unit 3 · Substitution

Worked example: substituting into fractions

$p = 3,\ q = 5,\ r = 3$. Work out

(a) $\dfrac{4p}{r}$ (b) $\dfrac{q^2 - 1}{p}$ (c) $\dfrac{4p + r}{q}$

(a)	**Step 1**	Work out the top	$4 \times 3 = 12$
	Step 2	Divide	$12 \div 3 = 4$
(b)	**Step 1**	Work out the top	$5^2 - 1 = 24$
	Step 2	Divide	$24 \div 3 = 8$
(c)	**Step 1**	Work out the top	$4 \times 3 + 3 = 15$
	Step 2	Divide	$15 \div 5 = 3$

Practice: substituting into fractions

4.1 $p = 3,\ q = 7,\ r = 2$. Work out

(a) $\dfrac{2p}{r}$ (b) $\dfrac{q^2 - 1}{p}$ (c) $\dfrac{4p + r}{q}$

4.2 $p = 5,\ q = 6,\ r = 4$. Work out

(a) $\dfrac{4p}{r}$ (b) $\dfrac{q^2 - 1}{p}$ (c) $\dfrac{4p + r}{q}$

4.3 $p = 7,\ q = 5,\ r = 2$. Work out

(a) $\dfrac{2p}{r}$ (b) $\dfrac{q^2 - 4}{p}$ (c) $\dfrac{4p + r}{q}$

4.4 $p = 4,\ q = 9,\ r = 2$. Work out

(a) $\dfrac{2p}{r}$ (b) $\dfrac{q^2 - 1}{p}$ (c) $\dfrac{4p + r}{q}$

4.5 $p = 8,\ q = 5,\ r = 3$. Work out

(a) $\dfrac{3p}{r}$ (b) $\dfrac{q^2 - 1}{p}$ (c) $\dfrac{4p + r}{q}$

4.6 $p = 9,\ q = 7,\ r = 6$. Work out

(a) $\dfrac{2p}{r}$ (b) $\dfrac{q^2 - 4}{p}$ (c) $\dfrac{4p + r}{q}$

Unit 3 · Expanding brackets

Worked example: expanding a single bracket

Expand

(a) $5(x+2)$ (b) $x(x+6)$ (c) $2k(3k+5)$ (d) Find the value of $2k(3k+5)$ when $k=3$

(a)	**Step 1**	Expand	$5\times x+5\times 2$
	Step 2	Answer	$5x+10$
(b)	**Step 1**	Expand	$x\times x+x\times 6$
	Step 2	Answer	x^2+6x
(c)	**Step 1**	Expand	$2k\times 3k+2k\times 5$
	Step 2	Answer	$6k^2+10k$
(d)	**Step 1**	Put in $k=3$	$2\times 3\times(3\times 3+5)$
	Step 2	Work it out	6×14
	Step 3	Answer	84

Practice: expanding a single bracket

5.1 Expand

(a) $4(x+3)$ (b) $x(x+4)$ (c) $3k(2k+4)$ (d) Find the value of $3k(2k+4)$ when $k=2$

5.2 Expand

(a) $6(x+2)$ (b) $x(x+3)$ (c) $4k(3k+1)$ (d) Find the value of $4k(3k+1)$ when $k=2$

5.3 Expand

(a) $3(x+7)$ (b) $x(x+8)$ (c) $5k(2k+3)$ (d) Find the value of $5k(2k+3)$ when $k=1$

5.4 Expand

(a) $7(x+4)$ (b) $x(x+5)$ (c) $2k(4k+7)$ (d) Find the value of $2k(4k+7)$ when $k=3$

5.5 Expand

(a) $8(x+5)$ (b) $x(x+6)$ (c) $6k(2k+1)$ (d) Find the value of $6k(2k+1)$ when $k=2$

5.6 Expand

(a) $9(x+1)$ (b) $x(x+2)$ (c) $3k(5k+2)$ (d) Find the value of $3k(5k+2)$ when $k=2$

Worked example: an expression for the area of a rectangle

A rectangle has a width of y cm. The length is 4 cm longer than the width.

(a) Write an expression for the length. (b) Show that the area is $A = y^2 + 4y$ (c) Work out the area when $y = 6$

(a)	Step 1	Longer: add 4	$y + 4$
	Step 2	Answer	$y + 4$ cm
(b)	Step 1	Area = length × width	$A = y(y + 4)$
	Step 2	Expand	$A = y^2 + 4y$
(c)	Step 1	Put in $y = 6$	$6^2 + 4 \times 6$
	Step 2	Work it out	$36 + 24$
	Step 3	Answer	60 cm^2

Practice: an expression for the area of a rectangle

6.1 A rectangle has a width of y cm. The length is 7 cm longer than the width.

(a) Write an expression for the length. (b) Show that the area is $A = y^2 + 7y$ (c) Work out the area when $y = 9$

6.2 A rectangle has a width of y cm. The length is 3 cm longer than the width.

(a) Write an expression for the length. (b) Show that the area is $A = y^2 + 3y$ (c) Work out the area when $y = 8$

6.3 A rectangle has a width of y cm. The length is 5 cm longer than the width.

(a) Write an expression for the length. (b) Show that the area is $A = y^2 + 5y$ (c) Work out the area when $y = 4$

6.4 A rectangle has a width of y cm. The length is 2 cm shorter than the width.

(a) Write an expression for the length. (b) Show that the area is $A = y^2 - 2y$ (c) Work out the area when $y = 10$

6.5 A rectangle has a width of y cm. The length is 9 cm longer than the width.

(a) Write an expression for the length. (b) Show that the area is $A = y^2 + 9y$ (c) Work out the area when $y = 3$

6.6 A rectangle has a width of y cm. The length is 6 cm shorter than the width.

(a) Write an expression for the length. (b) Show that the area is $A = y^2 - 6y$ (c) Work out the area when $y = 11$